

MULTIPARAMETRIC MRI AND SYNTHETIC DIFFUSION IMAGING IN THE CHARACTERISATION OF BREAST LESIONS

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BACKGROUND

Diffusion-Weighted Imaging (DWI) is an integral component of breast MRI, directly acquiring high b-value images (aDWI) suffers from low signal-to-noise ratio, prolonged echo times, and severe geometric distortion. Synthetic DWI (sDWI) via MAGIC software uses a mathematical monoexponential decay model to derive virtual high b-value images without acquisition penalties, but its progressive performance against standard aDWI800 on 3.0T MRI requires clinical validation.

OBJECTIVES

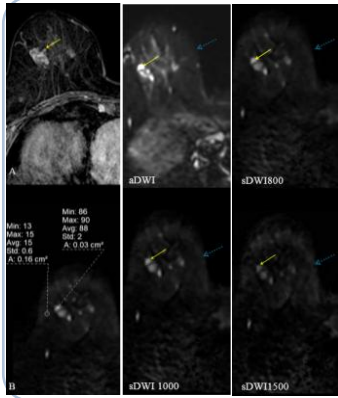
- To compare lesion conspicuity alongside qualitative and quantitative image quality across four b-value configurations (aDWI800, sDWI800, sDWI1000, and sDWI1500) on a 3.0T MRI.
- Secondarily, to evaluate the feasibility of quantitative image quality metrics—specifically signal-to-noise ratio (SNR), contrast-to-noise ratio (CNR), and contrast-to-parenchyma ratio (CPR)—for differentiating benign from malignant lesions. We also assessed the diagnostic accuracy of Apparent Diffusion Coefficient (ADC) values using continuous, data-derived optimal thresholds (<1.5×10⁻³mm²/s) and literature-based thresholds.

METHODOLOGY

Study Design: Single-centre prospective study at UMMC (Oct 2024–June 2026; Ethics ID: MREC 202492-14164).
Cohort: 23 patients with 35 histologically proven breast lesions (28 benign [80.0%], 7 malignant [20.0%]) undergoing 3.0T MRI.
MRI Protocol: 3.0T GE SIGNA Architect using 18-channel breast coil. Sequences included Axial T2 STIR, aDWI (b=0, 800s/mm²; TR 11947ms, NEX 8), 3D DCE-MRI, and sDWI (b800, b1000, b1500) reconstructed via GE MAGIC software.

Image Evaluation:
Qualitative: Consensus scoring by 2 radiologists on a 5-point Likert scale across 5 parameters.
Quantitative: Freehand ROI on tumour ("hottest spot") and normal parenchyma to derive SNR, CNR, and Cancer-to-Parenchyma Ratio (CPR).
Statistics: Non-parametric Friedman test, post-hoc Wilcoxon signed-rank tests (Holm-adjusted), ROC curve analysis, Youden's index, and logistic regression

RESULTS



Case Example:
A 69-year-old lady with ER-positive, PR-positive, and HER2 DCIS (ductal carcinoma in situ) in the right breast, 11 o'clock position.

A small freehand region of interest (ROI) was manually drawn within the identified tumour on the sDWI1000 image(B) , directly correlated with the DCE PH5 image (A) while excluding non-solid areas such as haemorrhage, necrosis, or cystic components. These ROIs were subsequently used to calculate the relative signal-to-noise ratio (SNR), contrast-to-noise ratio (CNR), and contrast-to-parenchyma ratio (CPR).

On aDWI800, the hyperintense lesion (yellow arrow) and adjacent tissue (blue dotted arrow) are visualised [SNR: 14.10, CNR: 13.50, CPR: 23.91]. sDWI800 improves background suppression while markedly increasing signal and contrast metrics over aDWI800 [SNR: 39.67, CNR: 32.06, CPR: 11.90]. sDWI1000 achieves peak lesion conspicuity with the highest SNR and CNR across all sequences [SNR: 44.00, CNR: 34.96, CPR: 5.87]. sDWI1500 offers maximum background suppression, though overall signal metrics decline [SNR: 28.67, CNR: 13.63, CPR: 4.30].

Qualitative Likert Scores & Conspicuity

Lesion Conspicuity: Peak median rating was achieved on intermediate sDWI (median 5 for sDWI800 and sDWI1000 vs 4 for aDWI800 and sDWI1500, $p = 0.00104$). Over 50% of lesions were rated higher on intermediate sDWI than on aDWI800.

Anatomical Clarity & Artifacts: aDWI800 was rated significantly superior to all sDWI sequences for normal parenchymal signal (median 4 vs 2 for sDWI1500, $p < 0.0001$), background fat suppression (4 vs 2, $p < 0.0001$), thoracic muscle/rib delineation (4 vs 2, $p < 0.0001$), and thoracic cage artifacts (3 vs 1, $p < 0.0001$)

Metric	aDWI800 Median (IQR)	sDWI800 Median (IQR)	sDWI1000 Median (IQR)	sDWI1500 Median (IQR)	$\chi^2(3)$	p-value
SNR	20.5 (7.96–32.0)	29.0 (13.42–45.92)	23.33 (11.12–44.67)	10.14 (6.0–15.67)	17.62	0.0005329
CNR	13.5 (5.55–26.77)	12.73 (6.92–20.83)	12.40 (5.61–21.54)	4.59 (3.13–8.59)	21.41	<0.000129
CPR	10.9 (4.71–14.5)	4.83 (3.0–8.60)	4.56 (3.65–7.62)	4.30 (2.09–7.53)	19.22	0.00025

Table 1: Quantitative Image Quality Comparison by DWI Sequence

sDWI1500 exhibited a severe drop in SNR and CNR compared to all lower b-value sequences ($p < 0.005$). aDWI800 achieved significantly superior CPR over all sDWI sequences ($p < 0.05$)

Key findings

Standard technical metrics (SNR, CNR, CPR) reflect sequence quality, not malignancy. The data-derived cutoff (ADC<1.5×10⁻³mm²/s) optimises sensitivity (85.7%) and NPV (93.8%), outperforming literature cutoffs.

Threshold (ADC ≤ 1.5 × 10 ⁻³ mm ² /s)	Sensitivity	Specificity	PPV	NPV	Accuracy
1.2 (Gullo et al., 2024)	42.90%	75.00%	30.00%	84.00%	68.60%
1.3 (Baltzer et al., 2020)	57.10%	71.40%	33.30%	87.00%	68.60%
1.5 (Youden-Optimal Data-Derived)	85.70%	53.60%	31.60%	93.80%	60.00%

Table 2: Diagnostic Performance of ADC & Cutoff Thresholds comparison

Lesion ADC Values: Malignant lesions demonstrated lower median ADC (1.3×10⁻³mm²/s) than benign lesions 1.7×10⁻³mm²/s, $p = 0.082$. Continuous ADC ROC AUC = **0.71742**

CONCLUSION / TAKE- HOME MESSAGE

1. Main finding

Intermediate synthetic DWI (sDWI800–1000) preserves high lesion conspicuity without scan-time penalties, whereas ultra-high weighting (sDWI1500) causes severe noise amplification and artifact degradation

2. Clinical meaning

Replacing std literature benchmarks with calibrated threshold (<1.5×10⁻³mm²/s) yields an exceptional 93.8%. Negative predictive value optimises non-invasive cancer exclusion and potentially prevents 15.6% of unnecessary biopsies.

3. Practice Recommendation

A balanced clinical protocol should pair aDWI800 for chest wall/anatomical evaluation with intermediate sDWI (b800–1000) for lesion characterisation

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QUICK POSTER CHECK

- ☐ Minimal text ☐ Results dominate ☐ Clear figures/tables ☐ One-sentence conclusion ☐ Readable fonts ☐ QR/contact optional